Name: ____________________ Date: ____________

Mixed Polynomial Factoring

This prehistoric monument on Salisbury Plain in England consists of a massive ring of standing stones, each weighing around 25 tons.

Match your answer in the solution bank to find the letter for each blank.

Factor each expression completely:

Factor completely:

(509) $x^2 + 7x + 10$

(142) $3x^2 + 15x$

(234) $x^2 - 36$

(612) $4x^2 - 12x$

(341) $x^2 - 49$

(815) $x^2 + 6x + 8$

(788) $x^2 + 8x + 15$

Solution Bank

Solution	*Letter*
$(x-6)^2$	[P]
$(x-7)(x+7)$	[H]
$3x(x+5)$	[S]
$(x-6)(x+6)$	[O]
$4x(x-3)$	[N]
$3x(x+15)$	[K]
$(x+2)(x+4)$	[T]
$(x+1)(x+10)$	[W]
$(x+2)(x+5)$	[E]
$(x+3)(x+5)$	[G]

___	___	___	___	___	___	___	___	___	___
142	815	234	612	509	341	509	612	788	509

Name: ______________________ Date: ______________

Algebra 1: Intermediate

Mixed Polynomial Factoring

These massive ancient geoglyphs in the Peruvian desert depict animals and plants, and are best viewed from the air.

Match your answer in the solution bank to find the letter for each blank.

Factor each expression completely:

Factor completely:

(405) $2x^2 - 50$

(166) $3x^2 - 15x + 18$

(918) $5x^2 - 20$

(892) $3x^2 - x - 2$

(239) $2x^2 - 5x - 3$

(782) $4x^2 + 12x - 40$

(314) $2x^2 + 16x + 30$

(527) $2x^2 + 7x + 3$

Solution Bank

Solution	*Letter*
$4(x+5)(x-2)$	[L]
$(2x+3)(x+1)$	[W]
$(3x+2)(x-1)$	[I]
$2(x-5)(x+5)$	[N]
$3(x-2)(x-3)$	[A]
$(2x+1)(x-3)$	[S]
$5(x-2)(x+2)$	[C]
$2(x^2-25)$	[M]
$3(x-1)(x-6)$	[P]
$2(x+3)(x+5)$	[E]
$(2x+1)(x+3)$	[Z]

___ ___ ___ ___ ___   ___ ___ ___ ___ ___
405 166 527 918 166   782 892 405 314 239

Name: ______________________ Date: ______________

Mixed Polynomial Factoring

This remote location in the Pacific Ocean is famous for its nearly 1,000 monumental stone statues, called moai, created by the early Rapa Nui people.

Match your answer in the solution bank to find the letter for each blank.

Factor each expression completely:

Factor completely:

(157) $25x^2 - 64y^2$

(349) $9x^2 - 100y^2$

(571) $9x^2 - 12x + 4$

(923) $4x^2 - 4x - 15$

(834) $-6x^2 - 2x + 4$

(718) $144x^2 - 121$

(481) $-6x^2 + 15x + 9$

(268) $6x^2 + 7x - 5$

(605) $-8x^2 + 12x + 20$

Solution Bank

Solution	*Letter*
$(12x - 11)(12x + 11)$	**[L]**
$(2x - 1)(3x + 5)$	**[I]**
$(2x + 3)(2x - 5)$	**[S]**
$(3x - 10y)(3x + 10y)$	**[T]**
$-3(2x - 1)(x + 3)$	**[B]**
$(3x - 2)^2$	**[D]**
$(2x + 1)(3x - 5)$	**[M]**
$-2(3x - 2)(x + 1)$	**[R]**
$(5x - 8y)(5x + 8y)$	**[A]**
$-3(2x + 1)(x - 3)$	**[E]**
$(5x - 8y)^2$	**[F]**
$-4(2x - 5)(x + 1)$	**[N]**

___ ___ ___ ___ ___ ___   ___ ___ ___ ___ ___ ___

481 157 923 349 481 834   268 923 718 157 605 571